Current view
training or user

Five-Stage SCD-GS Dataflow

Canonical update

Stage I
SIU
impact and confirmation

commit or confirmation

Stage II
FRU
tier directory and window state

Stage III
CIU
full, light, or filter issue

issue path

Stage IV
TRU
global tile-depth-gid drain

training trajectory

Stage V
TUU
residual, scalar feedback, Adam step

prior summary

metadata snapshot

prior summary and trace

render read

complete state + tier metadata

rendered pixel

Explicit State, Trace, and Metadata Storage

SplatSPM
prior summary and training trajectory

MetaSPM
FRU metadata snapshot

StateSPM
active, resident, candidate state and Adam vectors

FrameSPM
target and rendered pixels

target

target

state write

Candidate memory
64-byte aligned request and complete matching state fill

Target pixel

feedback canonical update